

SIMC 2026 — Declaration and Log of AI Assistance

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Endeavour Track · Team ID: [INSERT TEAM ID] · 27 May 2026

1. Declaration

This document is the companion AI-usage log required by the *Instructions to Participants 2026*. We used one AI assistant — **Claude Opus 4.7**, accessed through the Claude Code command-line tool — and only for clearly bounded, *non-decisional* support tasks. The substance of every solution — the mathematical modelling, the proofs, the choice of methods, the validation numbers, and the written interpretations — was produced and owned by the team. **Every AI suggestion was read, checked against our own working, edited where necessary, and re-executed by us before it was accepted into the project.** Nothing below was pasted in unread.

The entries that follow are a curated record of the recurring ways we used the assistant, each giving (i) the instruction we gave, (ii) a representative excerpt of what it returned, and (iii) exactly what we verified or changed before keeping it. The corresponding artifacts are in the submitted repository (src/, doc/, data/output/).

2. What AI was — and was not — used for

Used for (assistive, verified)	NOT used for (team only)
Drafting code comments & docstrings	Deriving the proofs (Q1, Q2)
Wording plot titles, axis labels, captions	Choosing the cleaning / clustering method
Plotting & L ^A T _E X boilerplate from our spec	Reading off any result or validation number
Re-checking our algebra for slips	The written interpretation of every result
Tightening report prose	The decision of what to claim and submit

3. Log of AI-assisted steps

Entry 1 — Internal code documentation (docstrings)

Prompt.

“Add a module-purpose docstring to the Task 1 validation cells. Explain the two-track check (direct iteration vs. closed form), state which variables each cell defines for later cells, and note the machine-precision assertion. Do not change any of the computation.”

AI output (excerpt, now in src/task1/task1.ipynb).

```
'''Purpose. Confirm the closed-form  $P(t)=c_1*v_1+c_2*(1-\epsilon)^t*v_2$  (1d) and its limit (1e)
by checking against a brute-force iteration that uses no analytical input.
Method: Track A iterates  $P(t+1)=M\Theta P(t)$ ; Track B evaluates the eigenmode formula.
Validation:  $\max|A-B| < 1e-10$  (asserted). Defines t, c1, c2, v1, v2, P_num.'''
```

Team verification.

We confirmed every claim in the docstring against the code (variable names, the $< 10^{-10}$ bound, the cell dependencies), re-ran the notebook end-to-end (all assertions pass), and kept the computation byte-for-byte unchanged. The docstring is description only.

Entry 2 — Plot labelling and captions

Prompt.

“Suggest concise axis labels, a title, and a one-sentence caption for the eigenvalue-spectrum figure (linear and log panels) that name the elbow at $k^* = 8$ and the noise floor. Maths in the labels should be valid $\text{\LaTeX}$.”

AI output (excerpt).

```
xlabel: "component index k" ylabel: r"eigenvalue  $\lambda_k$ "
title : "Eigenvalue spectrum (log scale)" axvline at k*=8 (label "elbow k* = 8")
caption: "Eight signal eigenvalues, then a flat noise floor: a sharp elbow at k*=8."
```

Team verification.

We supplied the value $k^* = 8$ (computed by our own code as the largest multiplicative drop $\lambda_8/\lambda_9 \approx 4.1$); the assistant only phrased the labels. We checked the figure renders and the numbers match our table before using it.

Entry 3 — Plotting infrastructure from our specification (the Q3g figure)

Prompt.

“We have covariance-space scores $\mathbf{Z}$ and Gram-space scores $\mathbf{Z}_g$ (both 924×4). Build a figure that proves they are the same point cloud up to a per-axis sign: a $|\text{correlation}|$ heatmap (4×4) plus the six PC-pair scatters overlaid, with the Gram scores sign-aligned to the covariance scores. Save to data/output/TASK 3/q3g_compare.png.”

AI output (excerpt, now cell 28 of src/task_3.ipynb).

```
signs = np.sign(np.diag(Mcorr)) # +-1 per matched component
Zg_al = Zg * signs # flip each Gram axis to match 3e
# left: imshow(|Mcorr|) heatmap ; right: 6 overlaid PC-pair scatters
print("max |Z - Zg(sign-aligned)| =", float(np.abs(Z - Zg_al).max())) # ~1e-14
```

Team verification.

The mathematical claim is ours (the SVD argument $\mathbf{Z} = \mathbf{U}\Sigma = \mathbf{Z}_g, \text{Q3g}$). We ran the cell and independently confirmed the diagnostics it prints — diagonal $|\text{corr}| = 1$, off-diagonal 0, and $|\mathbf{Z}| = |\mathbf{Z}_g|$ to $\approx 10^{-14}$ with PC1 and PC4 sign-flipped — before trusting the figure. The plot only *visualises* a result we had already proved.

Entry 4 — $\text{\LaTeX}$ report scaffolding and prose tightening

Prompt.

“Set up a 12pt Times, A4, single-spaced $\text{\LaTeX}$ skeleton for a ≤ 10 -page task-first report (executive summary, assumptions/notation, one section per task, exploratory, validation, references, AI-log). Then tighten our executive-summary paragraph to ≤ 8 lines without adding any new claim.”

AI output (excerpt).

```
\documentclass[12pt,a4paper]{article} \usepackage{mathptmx} ...
\section{Executive Summary} % one tight paragraph, names the headline number
```

Team verification.

We wrote the content; the assistant supplied formatting and condensed our wording. We compared the tightened paragraph line by line against our draft to confirm no claim was added, dropped, or altered, then compiled and proofread the PDF ourselves.

Entry 5 — Re-checking our algebra for slips

Prompt.

“Here is our Q1(g) result for a time-varying rate. Sanity-check the algebra: should the $\vec{v}_2$ factor be $(1 - \varepsilon(t))^t$ or the product $\prod_{s < t} (1 - \varepsilon(s))$? Explain which and why — do not rewrite our solution.”

AI output (excerpt).

“Because $\mathbb{M}(t)$ changes each step but shares the same ε -independent eigenvectors, the $\vec{v}_2$ mode picks up one factor $(1 - \varepsilon(s))$ *per step* — hence the ordered product, not the power. The power form would assume a single fixed matrix.”

Team verification.

This matched the correction we had already made in our notes (logged in `doc/FINAL/REPORT/changelog.md`, item M1). We then *proved* the product form ourselves and validated it numerically in `task1.ipynb` (the frozen residue $c_2\Pi_\infty$); the assistant’s reply was used only as a second pair of eyes, not as the source of the result.

Entry 6 — Numerical sanity checks we asked it to run

Prompt.

“Confirm two consistency checks on our covariance computation: (a) $\text{tr}C$ equals $\sum_i \lambda_i$; (b) the top-8 Gram eigenvalues equal the top-8 covariance eigenvalues. Report the numbers, do not interpret them.”

AI output (excerpt).

```
sum(eigenvalues) = 7.9007 vs tr(C) = 7.9007
gram-vs-cov top-8 match (np.allclose, atol=1e-6): True
```

Team verification.

We re-ran both checks ourselves from `src/task_3.ipynb` and reproduced the same numbers. These are invariance checks (rotation invariance of the trace; the Q2(d) shared-spectrum theorem) that we use as evidence in the report; the interpretation in the report is entirely our own.

4. Integrity statement

We affirm that the analytical reasoning, the modelling and method choices, the numerical results, and the written interpretations in our submission are the team’s own work. AI assistance was limited to documentation, presentation, plotting/L^AT_EX boilerplate, prose editing, and second-pass checking of work we had already done, as logged above. No result was reported that we did not independently reproduce, and no text was submitted that we did not review and edit.